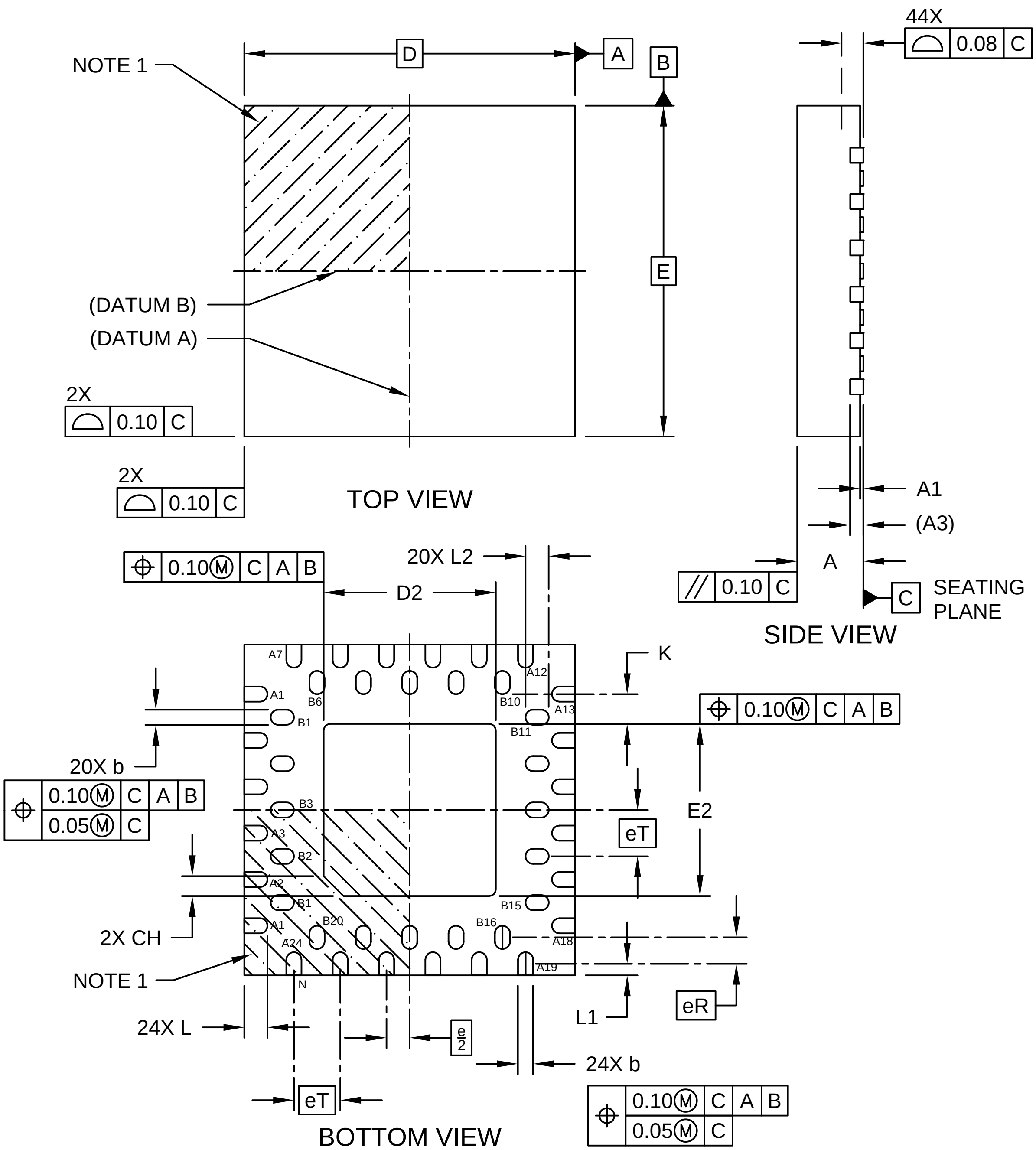
44-Lead Very Thin Plastic Quad Flat, No Lead Package (S3B) - 5x5x1.0 mm Body [VQFN] Dual Row; Atmel Legacy Global Package Code ZDS

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



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44-Lead Very Thin Plastic Quad Flat, No Lead Package (S3B) - 5x5x1.0 mm Body [VQFN] Dual Row; Atmel Legacy Global Package Code ZDS

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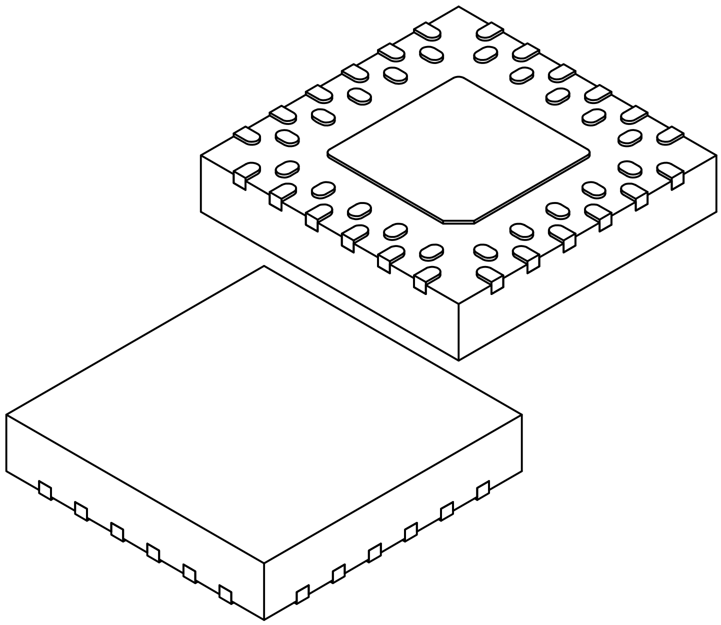

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N		44	
Pitch	eT		0.70 BSC	
Outer Row Center to Inner Row Center	eR		0.40 BSC	
Overall Height	A	0.80	0.90	1.00
Standoff	A1	0.00	0.02	0.05
Terminal Thickness	A3		0.20 REF	
Overall Length	D		5.00 BSC	
Exposed Pad Length	D2	2.55	2.60	2.65
Overall Width	E		5.00 BSC	
Exposed Pad Width	E2	2.55	2.60	2.65
Terminal Width	b	0.18	0.23	0.30
Outer Row Terminal Length	L	0.30	0.35	0.40
Edge to Terminal Center	L1		0.18 BSC	
Inner Row Terminal Length	L2	0.30	0.35	0.40
Terminal to Exposed-Pad	K	0.40	-	-
Exposed Pad Index Chamfer	CH	-	0.30	-

Notes:

1. Pin 1 visual index feature may vary, but must be located within the hatched area.
2. Package is saw singulated
3. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
   REF: Reference Dimension, usually without tolerance, for information purposes only.

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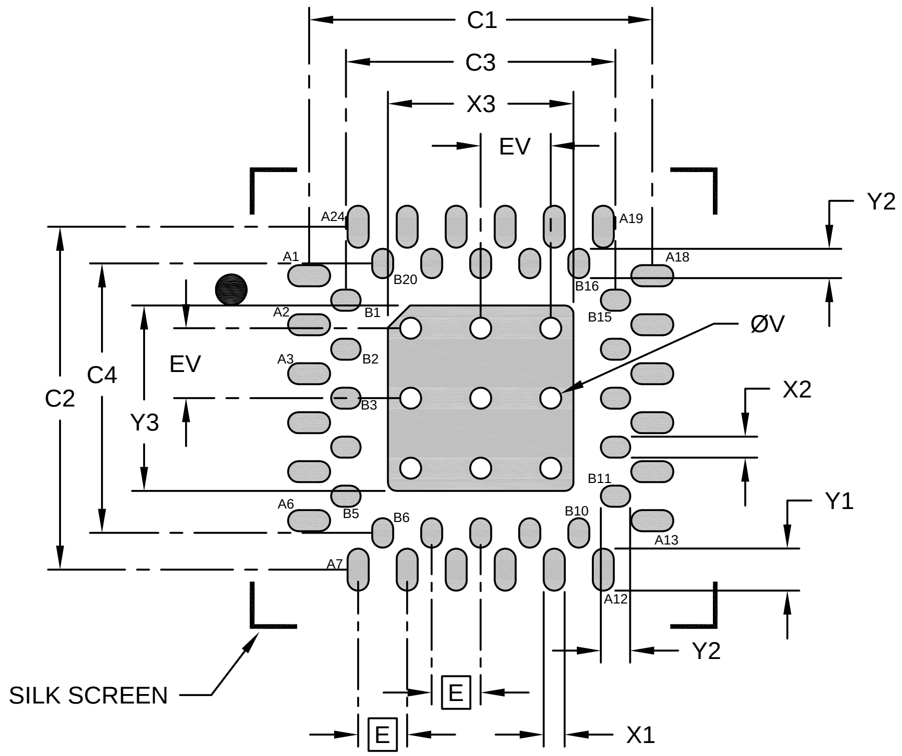
44-Lead Very Thin Plastic Quad Flat, No Lead Package (S3B) - 5x5x1.0 mm Body [VQFN] Dual Row; Atmel Legacy Global Package Code ZDS

Note:	For the most current package drawings, please see the Microchip Packaging Specification located at http://www.microchip.com/packaging



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E		0.70 BSC	
Optional Center Pad Width	X3			2.65
Optional Center Pad Length	Y3			2.65
Outer Contact Pad Spacing	C1		4.90	
Outer Contact Pad Spacing	C2		4.90	
Inner Contact Pad Spacing	C3		3.85	
Inner Contact Pad Spacing	C4		3.85	
Outer Contact Pad Width (X24)	X1			0.30
Outer Contact Pad Length (X24)	Y1			0.60
Inner Contact Pad Width (X20)	X2			0.30
Inner Contact Pad Length (X20)	Y2			0.42
Thermal Via Diameter	V		0.30	
Thermal Via Pitch	EV		1.00	

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M
   BSC: Basic Dimension. Theoretically exact value shown without tolerances.
2. For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

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